

Contactless High Dynamic-Range Blood Pressure Estimation by Pulse Morphological Features Based on Accurate Doppler Radar Detection

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Abstract—Blood pressure (BP) can be estimated contactlessly based on Doppler wrist pulse (DWP) detection. However, this technique faces the challenge of interferences from the respiratory and its harmonic components, which can limit the accuracy of the BP estimation. This article first presents an adaptive respiratory harmonic filtering (ARHF) technique, aiming to mitigate the interference of respiration on the wrist pulse signal. Also, a customized 24-GHz continuous wave radar system with high linearity and a high gain antenna is used for better DWP detection. Second, a novel radar-based BP estimation model based on multidimensional morphological features extraction is proposed for high-accuracy and high-robustness BP estimation. In a stationary BP estimation experiment with eight subjects over three consecutive days, the proposed method demonstrates an average mean error and standard deviation of 2.60 ± 7.89 mmHg and 1.01 ± 2.84 mmHg in the sitting posture for systolic and diastolic blood pressure (SBP and DBP), respectively, achieving the highest estimation accuracy among the recent contactless radar-based BP detection. Additionally, a specific BP fluctuation range experiment is conducted to continuously measure BP, including varying BP fluctuations across five stages. In the worst case scenario, the proposed method exhibits SBP and DBP errors

of 2.50 ± 2.98 and 8.34 ± 9.86 mmHg, respectively, affirming the robustness of the proposed technique. The experimental findings highlight the feasibility and reliability of the DWP extraction and subsequent contactless BP estimation techniques in the healthcare domain.

Index Terms—Adaptive respiratory harmonic filter (ARHF), contactless blood pressure estimation, diameter variation ratio (DVR), Doppler radar, Doppler wrist pulse (DWP).

I. INTRODUCTION

HYPERTENSION plays a key role as a risk index for cardiovascular disease (CVD). Regular blood pressure (BP) monitoring is crucial for early CVD detection and prevention as the population ages. However, traditional cuff-based sphygmomanometers, leading to discomfort, are not conducive to continuous monitoring. To overcome the limitations of conventional BP measurement, much research has focused on developing continuous BP measurement methods based on pulse transit time (PTT) [1], [2], [3], [4], [5], [6], among which, the utilization of wrist pulse signals for BP estimation has become a highly concerning field [7], [8], [9], [10]. Wrist pulse, a physiological signal reflecting the parameters related to blood vessels, originates from cardiac activity. Alterations in several characteristic points of the wrist pulse wave can indicate heart or blood vessel parameter changes [11], [12], [13]. Thus, similar to the electrocardiogram (ECG) [14], [15] or the Doppler Cardiogram [16], [17], [18], wrist pulse wave features can serve as an indicator of CVD [19], which can also be used for BP estimation.

Many studies use various sensors to capture the wrist pulse waveform for subsequent feature extraction in continuous BP estimation, such as cameras [1], [2], contact photoplethysmography (PPG) sensors [3], [4], [5], [6], piezoelectric sensors [7], or radars [8], [9], [10]. However, the camera-based approach requires additional light and also brings privacy concerns. The contact sensor faces challenges of long-term wear due to mandatory portability and issues related to battery consumption. Radar is less susceptible to light interference and can measure around the clock. Nevertheless, previous multiradar-based methods [9], [10] suffer from complexity in synchronization. Therefore, employing a single radar is a more practical solution.

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When using a single radar, respiration and its harmonics can increase the positioning error of wrist pulse waveform feature points [20], leading to inaccuracies in feature point extraction and consequently increasing errors in BP estimation. To mitigate this problem, a prevalent technique is to use a finite impulse response (FIR) filter for linear filtering [21]. However, this method may not effectively attenuate the impact of the respiratory harmonics [17]. Certain methods, like employing the infinite impulse response (IIR) filter in [8], may cause phase imbalance and affect the system's linearity. This can pose challenges for wrist pulse wave detection. Many studies focusing on a single radar for BP estimation [8], [22] often lack comprehensive discussions on respiratory harmonic suppression, resulting in interference affecting related features. There remains considerable potential for enhancing the accuracy of blood pressure estimation. Additionally, there is a lack of experimental validation concerning the model's performance when faced with substantial BP variations, highlighting the necessity to assess the method's robustness. Therefore, to achieve precise wrist pulse wave data, this article proposes a high-linearity signal processing method, called the adaptive respiratory harmonic filtering (ARHF) method, to effectively suppress respiratory harmonics, alongside experiments evaluating the robustness in the presence of BP fluctuations.

In contactless BP estimation methods, PTT and pulse wave velocity (PWV) serve as prominent features [1], [5], [6], [8], [9], [10], [23]. PTT is usually defined as the duration of the arterial pulse propagation between two arterial points, such as from the R wave of ECG to the peak of the PPG signal. Nevertheless, acquiring data from multiple arterial points requires deploying multiple sensors, leading to a high system complexity.

Recently, pulse wave detection using a single radar at the wrist to obtain reflective pulse transit time (rPTT) has shown some effectiveness in blood pressure estimation [22]. From a physiological perspective, blood pressure is primarily influenced by factors such as arterial compliance, peripheral resistance, and cardiac output [5]. Methods that estimate blood pressure based solely on PTT-based parameters, which reflect only arterial compliance, exhibit limited accuracy due to incomplete representation.

To address this concern, this article proposes a single-radar-based acquisition system, which effectively mitigates the complexity arising from the use of multiple sensors. Meanwhile, leveraging the precise Doppler wrist pulse (DWP) signal acquired by radar, this study introduces the novel BP estimation model based on multidimensional morphological features extraction, which combines the changing characteristics of the time dimension and scale dimension of pulse wave activity, for accurate dynamic-range BP estimation.

In this article, a single-radar-based DWP and blood pressure estimation method is proposed. The ARHF method is introduced to improve the DWP extraction accuracy, and a 24-GHz continuous wave (CW) Doppler radar system (DRS), which has high linearity and reliability in detecting object vibrations, is implemented. Initially, the method is evaluated through simulation experiments to address the effect of respiration on wrist pulse waves. Simulation results demonstrate

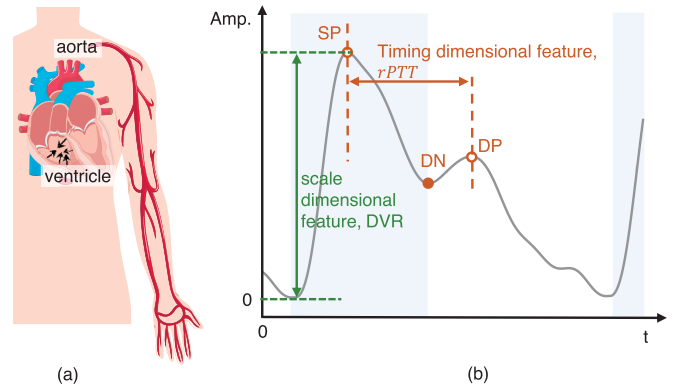


Fig. 1. (a) Mechanism illustration of wrist pulse wave generation, with dark arrows representing the contraction direction. (b) Wrist pulse wave and features used to estimate the BP, respectively. The light blue region represents the systolic period. SP: systolic peak; DN: diastolic notch; DP: diastolic peak.

that the proposed method can effectively extract wrist pulse waves while suppressing respiratory harmonics. Compared to the ground truth, the radar system accurately detects DWP signals with a small time difference. Furthermore, a novel BP estimation model is proposed in this article based on multidimensional morphological feature extraction. Experimental results over three consecutive days reveal improved accuracy of the proposed method compared to the current contactless radar-based approach. The total mean error (ME) and standard deviation (SD) for systolic and diastolic blood pressure (SBP and DBP) in the sitting posture are 2.60 ± 7.89 and 1.01 ± 2.84 mmHg, respectively. In the lying posture, the total ME and SD for SBP and DBP are 4.96 ± 10.15 and 1.53 ± 4.68 mmHg, respectively. An additional robustness experiment estimating BP variations was conducted through five distinct stages, characterized by short-term BP fluctuations. These experimental results further affirm the precision and robustness of the proposed method. These findings highlight the potential of employing DRS for precise CVD analysis in diverse medical environments such as clinics and nursing homes. The contribution of our work can be summarized as follows.

- 1) A contactless high dynamic-range BP estimation model based on multidimensional morphological features extraction is proposed. This method outperforms the existing radar-based approaches by providing precise SBP and DBP measurements. Meanwhile, a specific BP fluctuation range experiment is conducted to verify the robustness of the proposed method.
- 2) The ARHF method is introduced to extract the DWP signals. Both simulation and experiment demonstrate its effectiveness in suppressing respiratory and its harmonics.
- 3) A 24-GHz CW DRS is customized to ensure accurate detection of wrist motion. The system has high linearity and can detect amplitudes as tiny as $2 \mu\text{m}$.

II. THEORY

A. Wrist Pulse Wave

Fig. 1(a) demonstrates the ventricle contraction process. During blood circulation, when the ventricles contract, the aortic valve opens, and blood is pumped into the aorta. The

wrist pulse waveform consists of an ascending and a descending phase. The ascending phase indicates the abrupt arterial expansion during ventricular contraction, while the descending phase represents ventricular relaxation. The ascending phase rises rapidly and smoothly, representing rapid ventricular ejection, increased aortic blood volume, and an elevation in BP.

The closure of the aorta signifies the start of diastole following ventricular contraction [12], [24], [25]. Currently, blood within the aorta tends to retrograde, resulting in a notch on the rapidly descending branch of the wrist pulse wave at the aortic valve. Subsequently, the blood countercurrent hits the aortic valve, unable to enter the ventricle due to its tight closure. Consequently, it retreats to the aorta, leading to a rapid rise in arterial pressure following a sudden drop. This creates an upward wavelet after the notch of the arterial pressure, which is known as the dicotric notch [12]. The wrist pulse wave then continues to decline as the blood continues to move forward. Fig. 1(b) illustrates several characteristic points of the pulse wave signal, including the systolic peak, dicotric notch, diastolic peak, rPTT, and the diameter variation ratio (DVR), which are used for BP estimation.

B. DWP Detection Based on Radar

The Windkessel model is a commonly used medical model for analyzing blood vessels and BP in arteries, considering the relationship between blood flow and BP. It can be expressed as [26]

$$F(t) = \frac{P(t)}{Z} + C \frac{dP(t)}{dt} \quad (1)$$

where $F(t)$ is the flow of blood, $P(t)$ is the arterial BP, Z is the peripheral resistance, and C is the arterial compliance. Meanwhile, in the cylindrical coordinate system of the arterial wall, the force balance of the arterial wall needs to meet the following conditions [27]:

$$\frac{d\sigma_{rr}}{dr} + \frac{1}{r}(\sigma_{rr} - \sigma_{\theta\theta}) = 0 \quad (2)$$

where r is the radius, σ_{rr} and $\sigma_{\theta\theta}$ are the stress in the radial and circumferential directions, respectively. According to the hyperelastic model for the human artery [27], the strands can be represented by changes in the internal area of the artery. $P(t)$ can be obtained by integrating force balance (2) from the inner radius $r = RR$ to the deformed outer radius $r = RR + h$

$$P(t) = \int_{-P}^0 d\sigma_{rr} = \int_{RR}^{RR+h} \frac{1}{r}(\sigma_{\theta\theta} - \sigma_{rr})dr. \quad (3)$$

These relationships demonstrate that BP directly influences the cross section of arteries. Specifically, the walls of blood vessels are affected, leading them to contract and expand. This process generates faint vibrations on the skin's surface. These subtle vibrations can potentially be detected through Doppler radar technology, offering a basis for noninvasive blood pressure monitoring.

Fig. 2 shows how Doppler radar detects DWP. After emitting the electromagnetic waves, the received in-phase and quadrature (I/Q) signal from the radar can be written as follows

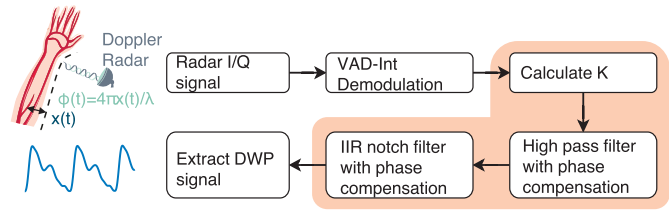


Fig. 2. Flowchart of how the radar extracts DWP signal. K is the respiratory and harmonics index. The light orange color indicates the contribution of this article.

assuming the movement of the wrist skin is a single-frequency movement:

$$\begin{aligned} IF_I[t] &= E_I \cos\left(\theta_0 + \frac{4\pi x[t]}{\lambda} + \Delta\phi\right) + dc_I \\ IF_Q[t] &= E_Q \sin\left(\theta_0 + \frac{4\pi x[t]}{\lambda} + \Delta\phi\right) + dc_Q \end{aligned} \quad (4)$$

where E_I and E_Q are the magnitudes of the I/Q signal, respectively; $x[t]$ denotes the pulse skin motion; λ represents the carrier wavelength, θ_0 is the initial phase noise, $\Delta\phi$ stands for residual phase noise, and the dc_I and dc_Q are the dc offset of each channel. The vectors analytic demodulation integration (VAD-Int) [18], proven for its high accuracy and robustness against noise for linear reconstruction in [18], is applied on (4) to linearly extract the wrist motion, which is the phase information in the IF signal.

The radar phase signal primarily carries the vibration information of the skin surface. The detected signal $x[t]$ contains not only the vibration of the wrist pulse itself but also the breath and its harmonic components. This presents challenges for the extraction of DWP.

C. Adaptive Respiratory Harmonic Filtering Method

To obtain an accurate wrist pulse wave signal, it is important to exclude the harmonic component of respiration from the final signal. The ARHF method is proposed to obtain the DWP by linear demodulation from the radar. This article proposes the respiratory and harmonics index K as

$$\lfloor K \rfloor = \frac{f_n}{f_r} \quad (5)$$

where f_r is the main respiration frequency and f_n represents the threshold frequency. $\lfloor \cdot \rfloor$ is the floor function, where the output is the greatest integer less than or equal to the input variable. In this work, the value of f_n is set to 1 Hz. Through experimentation, it was found that in the range greater than 1 Hz, the respiration harmonics energy is significantly less than the wrist pulse component, and the impact on the wrist pulse is almost negligible [20], [28]. The intensity of respiration mainly causes the drift of the overall detection waveform, so only the influence of low-frequency components on the wrist pulse wave is considered.

The DWP extraction process is illustrated in Fig. 2. First, linear demodulation is performed to obtain the amplitude of the wrist motion. Second, the respiratory and harmonics index K is calculated by (5), and an IIR high-pass filter with phase compensation method is applied. An IIR notch filter

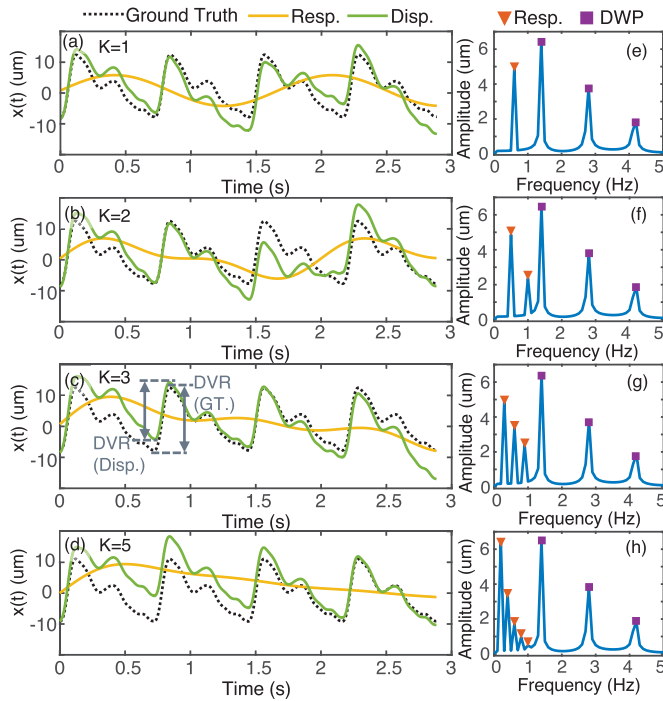


Fig. 3. Simulation results for the DWP combined with respiratory and its harmonic with different K values. (a)–(d) Original ($x(t)$) waveform with the cases $K = 1, 2, 3, 5$, respectively. (e)–(h) Corresponding spectrum results annotated with respiratory, respiratory harmonics, and Doppler wrist pulse (DWP) components. The black rectangle is the zoomed-in region to show the influence of the respiratory and its harmonics on the pulse signal. Resp.: respiratory signal; Disp.: displacement signal; GT.: ground truth.

is subsequently applied to eliminate the influence of respiratory harmonics, and in combination with phase compensation methods, the DWP is acquired.

When processing the demodulated signal, the first step is to perform IIR filtering on the demodulation, with a cutoff frequency of 0.7 Hz and a transition band of 0.1 Hz [29]. Subsequently, the phase compensation algorithm [30] is applied to compensate for the phase offset caused by the IIR filter. One of the main reasons for using the IIR filter is that compared with the FIR filter, the IIR filter only needs lower order and less computation time to achieve almost the same effect, which is of great significance in reducing the computation complexity [31]. Next, based on the previously calculated coefficient K , an IIR notch filter with a bandwidth of 0.002 Hz is employed to eliminate the harmonic components and minimize the impact on the DWP. Finally, after processing, the phase compensation method is reapplied to rectify the phase shift caused by the IIR filter.

To verify the effectiveness of the proposed method, a simulation experiment is performed. The components brought by breathing and the waveform of the wrist pulse wave are simulated. The frequency of the wrist pulse wave is set to 1.4 Hz, and the frequency of breathing is set to different values to test the effectiveness of the proposed method under different K values, which are 0.6, 0.5, 0.3, and 0.2 Hz, respectively.

Fig. 3 shows the respiration waveform (Resp.) under different conditions, the synthetic wrist pulse displacement (Disp.), and the corresponding frequency domain distribution. When

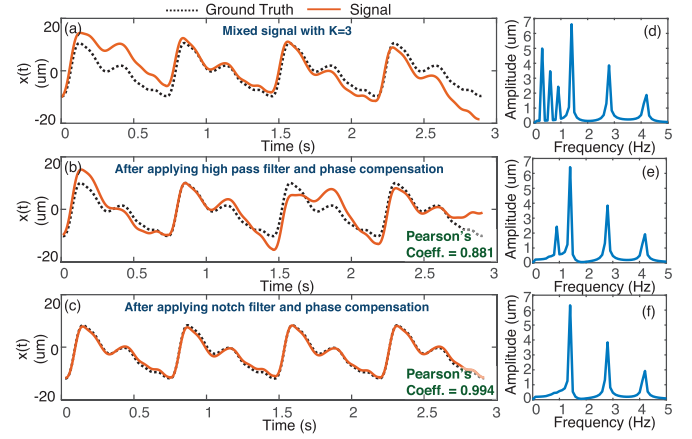


Fig. 4. Process when applying ARHF method with respiratory frequency is 0.3 Hz and $K = 3$. (a)–(c) Original ($x(t)$) and ground-truth waveform after each step. (d)–(f) Corresponding spectrum results. Pearson's coefficient is calculated between the signal and the ground truth. The black rectangle is the zoomed-in region to show the influence of the respiratory harmonics on the pulse signal. Pearson's Coeff.: Pearson's coefficient.

the coefficient of K is 1, that is, there is only the respiration component and no harmonic component, which means that the respiration component can be well removed by the high-pass filter at the beginning. However, when K is not equal to 1, it means that at least one respiration harmonic component will not be filtered out by the high-pass filter, so it is necessary to use the proposed method to reduce the influences. From the preliminary displacement simulation results under different respiratory states in Fig. 3, it can be observed that without filtering respiratory motion, the extraction of the DVR parameter proposed in this article would be directly impacted. As demonstrated in Fig. 3(c), the DVR values extracted from the second period of the displacement (Disp.) and ground truth (GT.) signal show considerable deviation due to respiratory interference. This observation emphasizes the necessity of effectively filtering respiratory and its harmonics to ensure accurate extraction of the proposed model parameters.

Fig. 4 shows the process of the proposed ARHF method when the dominant respiratory frequency is 0.3 Hz and $K = 3$. The whole process can be summarized as follows.

- 1) *Step 1 (High-Pass Filtering and Phase Compensation)*: Apply a high-pass filter with the phase compensation method to remove the low-frequency component. The result and the corresponding spectrum can be found in Fig. 4(b) and (e).
- 2) *Step 2 (Notch Filtering and Phase Compensation)*: Design a notch filter centered at 0.9 Hz in this case with 0.002 Hz bandwidth. The final result shows the effective removal of respiratory harmonics while preserving the desired signal components.

It can be seen that the output signal after the proposed method can obtain a Pearson's coefficient of 0.99 compared with the ground truth, which means that the proposed method can well suppress the influence of the respiratory harmonic band and recover the accurate wrist pulse wave signal.

To compare the effectiveness of the proposed method, the commonly used filtering methods, such as the IIR filter with

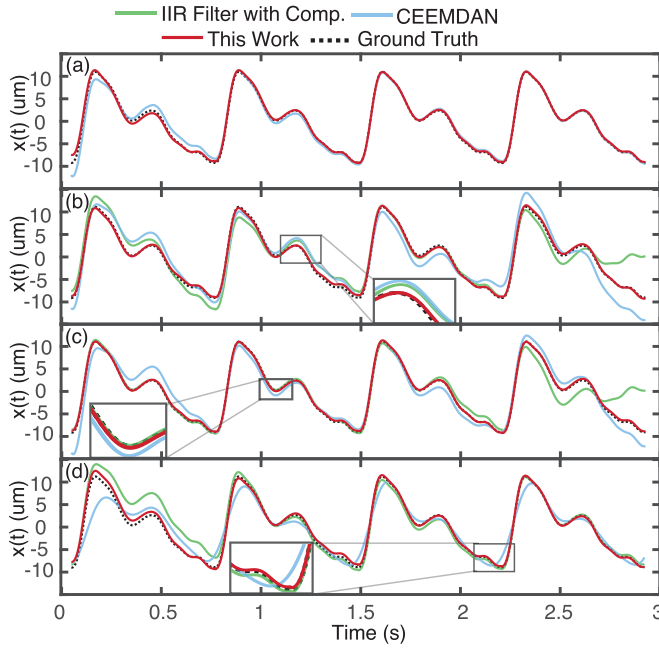


Fig. 5. Comparison among other filtering methods with different K . (a)–(d) Represents original waveforms ($x(t)$) when $K = 1, 2, 3, 5$, respectively. The black rectangle is the zoomed-in region to show the difference among all filtering methods. Comp.: Compensation.

compensation and the complete ensemble empirical mode decomposition (CEEMDAN) [32], are also used for comparison. The comparison result is shown in Fig. 5. When $K = 1$, there is no harmonic component. At this time, all methods can achieve good results. When $K = 2$, there is a component with respiratory harmonics of 1 Hz. Only using a high-pass filter cannot suppress these harmonic components well, resulting in a certain degree of signal drift. When the acquired data contains the respiratory harmonic, the CEEMDAN method has a large distortion in most cases. In practice, the existence of respiratory harmonics will make the CEEMDAN method that cannot effectively obtain the DWP.

The respiratory filtering simulation results in Fig. 5 show that different respiratory filtering algorithms affect the extraction of the DWP to varying degrees, especially in terms of the positioning of the two characteristic peaks of the DWP. For ease of observation, the region around the DWP characteristic waveform is highlighted within the black frame in Fig. 5. Compared to the ground truth, it can be seen that other respiratory filtering algorithms cause a certain degree of deviation in the positions of the characteristic peaks, while the ARHF proposed in this article preserves the characteristics of the DWP relatively well. This feature ensures the accurate extraction of the features used in the subsequent BP estimation algorithm proposed in this article, thereby improving the accuracy of BP estimation.

D. Noncontact Blood Pressure Estimation Based on Morphological Waveform Features

Based on the physiological principles of blood pressure calculation, the PTT parameter primarily reflects arterial compliance, while a key factor in peripheral resistance is mainly

related to the radius of the blood vessels [5]. The basis for the mechanism of peripheral vascular resistance is expressed by the Hagen–Poiseuille equation [33]

$$R = \frac{8Ln}{\pi d^4} \quad (6)$$

where R is the peripheral resistance, L is the length of the vessel, n is the viscosity of the blood, and d is the diameter of the blood vessel.

The novelty proposed blood pressure estimation model incorporates the DVR parameter, which describes changes in vessel diameter (Δd). DVR can be obtained through precise linear detection of wrist pulse displacements using radar technology. DVR is defined as

$$\text{DVR} = e^{-\alpha(\Delta d)^4} \quad (7)$$

where α represents the constant of different human wrist muscle tissues, which is related to the arterial compliance C and is constant in a relatively short period.

Based on the Windkessel model [34], the diastolic blood pressure (DBP) can be theoretically expressed as

$$\text{DBP} = P_0 \cdot e^{-t/RC} \quad (8)$$

where P_0 is the end-systolic aortic pressure. Based on (6), the relationship between DBP and DVR can be derived as follows:

$$\text{DBP} \propto \text{DVR}. \quad (9)$$

Therefore, DBP can be derived with calibrated DBP_0 and DVR_0

$$\text{DBP} = \text{DBP}_0 \times \frac{\text{DVR}}{\text{DVR}_0}. \quad (10)$$

Subsequently, as claimed by the Bramwell–Hill equation [35]

$$\text{PWV} = \sqrt{\frac{V_b \Delta P}{\rho \Delta V}} \quad (11)$$

where ΔP is calculated by subtracting the systolic blood pressure (SBP) from the DBP, and it can be named the pulse pressure (PP). The V_b is the blood volume. The PP can be written as

$$\text{PP} = \text{PP}_0 \times \left(\frac{\text{rPTT}_0}{\text{rPTT}} \right)^2 \quad (12)$$

where PP_0 and rPTT_0 are the first measurements for further calibration.

Finally

$$\begin{aligned} \text{SBP} &= \text{PP} + \text{DBP} \\ &= \text{PP}_0 \times \left(\frac{\text{rPTT}_0}{\text{rPTT}} \right)^2 + \text{DBP}_0 \times \frac{\text{DVR}}{\text{DVR}_0}. \end{aligned} \quad (13)$$

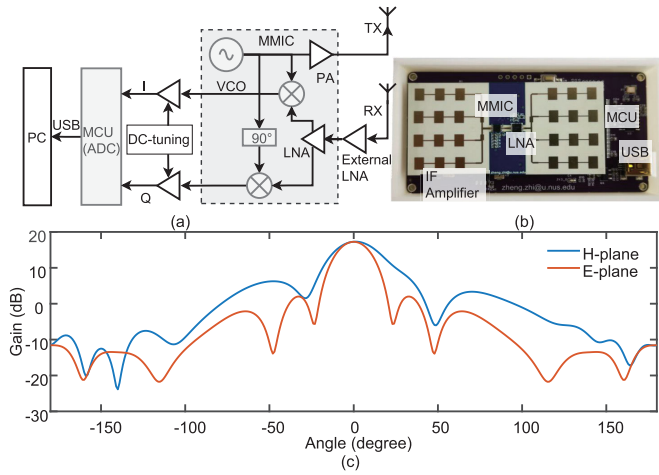


Fig. 6. (a) Schematic of the customized Doppler radar system. (b) Top-view photographs of the radar system. (c) Radiation pattern of the customized radar's antenna.

III. EXPERIMENT AND RESULTS

A. Customized Radar System

Fig. 6(a) illustrates the schematic of the customized radar system. First, the radar signal containing wrist motion information is received by the antenna, and then the signal is amplified by a K-band low-noise amplifier, LNA024 (Indie Semiconductor, CA, USA), enhancing the system's SNR. Subsequently, the signal will be converted into baseband I/Q signals through a 24-GHz silicon germanium millimeter-wave integrated circuit, BGT24LTR11 (Infineon Technologies, Germany). A dc tuning circuit is adopted to satisfy the input range of a single-supply analog-to-digital converter (ADC) integrated inside the microcontroller unit (MCU). The I/Q signals are digitized by ADC with a 1000-Hz sampling rate and then transmitted to the PC via a USB cable. The hardware of the developed radar is depicted in Fig. 6(b), where the substrate, ROGERS 4350b, has a thickness of 10 mils and a dielectric constant of 3.48. The antenna is designed with a gain of 17.3 dBi. The radiation pattern of the antenna is illustrated in Fig. 6(c), with E-plane and H-plane 3-dB beamwidths of 19.8° and 26.6°, respectively.

B. Radar Performance

A pendulum experiment was conducted to verify the radar system's linearity and sensitivity before proceeding with further experiments. The pendulum motion exhibits single-frequency periodic behavior [36], and measuring the oscillation frequency of the pendulum using radar can verify the radar system's linearity. For a simple pendulum motion with a small initial swing angle ($<10^\circ$), the simple harmonic oscillation frequency remains independent of both the amplitude and the pendulum ball mass. The oscillation frequency of the pendulum f_p can be estimated using the formula below when the suspension string length l is known

$$f_p = \frac{1}{2\pi} \sqrt{\frac{g}{l}} \quad (14)$$

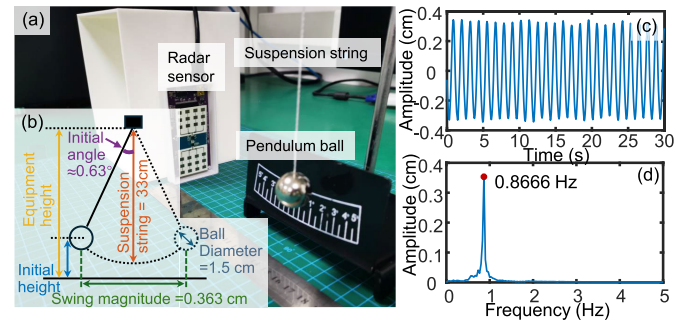


Fig. 7. (a) Experiment setup of the pendulum ball. (b) Schematic illustrating the relationship of the parameters. (c) Demodulated motion. (d) Corresponding frequency spectrum.

where g is the local gravitational acceleration. Such a harmonic motion $x_h(t)$ can be analytically expressed as

$$x_h(t) = l\theta \cos(2\pi f_p t) \quad (15)$$

where θ is the initial swing angle. In the experiment, the pendulum device was positioned 15 cm away from the radar, comprising a 1.5-cm diameter ball with about 3.9 cm initial height, corresponding to the initial swing angle of 0.63° and a suspension string of length of 33 cm, as shown in Fig. 7(a). Fig. 7(b) illustrates the relationship among the parameters in the pendulum experiment. Based on (14) and (15), the theoretical oscillation frequency and magnitude were calculated as 0.867 Hz and 0.363 cm, respectively. Fig. 7(c) and (d) demonstrates the motion waveform demodulated by a 30-s radar alongside its corresponding frequency spectrum. The magnitude detected by the radar of the pendulum ball is 0.364 cm, which is consistent with the theoretical calculation result. This experiment has preliminarily verified the accuracy of detecting the pendulum displacement. The spectrum result shows a close match between the frequency of the demodulated waveform and the calculated frequency, displaying pure frequency-domain components without any harmonics. This illustrates the high linearity of the proposed system.

Subsequently, the suspension string length was adjusted to evaluate the sensitivity of the proposed radar system in a short period. The length was shortened to 4.8 cm with an equivalent initial swing angle of 1.15° during the experiment. After the adjustment, the oscillation frequency increases, which leads to a rapid decrease in amplitude, and the amplitude will eventually be drowned by noise. Fig. 8 displays the reconstructed radar waveform over 90 s, demonstrating a gradual reduction in the pendulum's amplitude until obscurity by noise. The initial swing displacement can also be obtained according to (15), and the radar demodulation result is also consistent with the theoretical calculation result. Additionally, five blocks were magnified, displaying their corresponding magnitudes and frequencies. By using a 10-dB SNR threshold [16], the radar system proposed in this article can detect amplitudes as small as $2 \mu\text{m}$.

C. Different Scenarios for Wrist Pulse Measurement

Two different scenarios were conducted in this experiment to verify the accuracy of DWP. One is the tester with the sitting

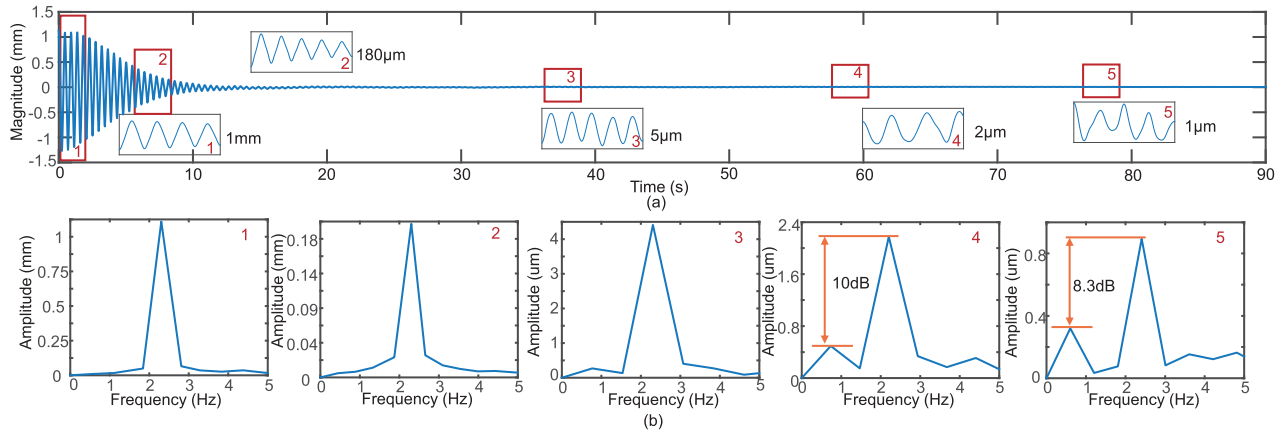


Fig. 8. Pendulum experimental results. (a) Demodulated radar signal. (b) Corresponding frequency spectrum of five blocks.

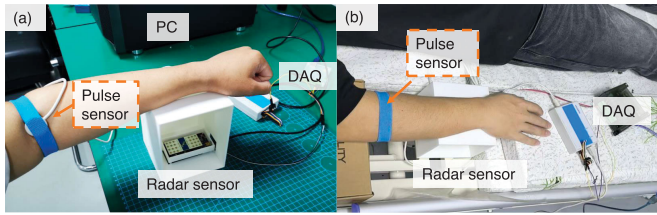


Fig. 9. Experiment setup for two different scenarios. (a) Sitting position. (b) Lying position. DAQ: Data acquisition board.

posture and the other is the tester with the lying posture. The experimental scenarios are shown in Fig. 9. The setting of these two experiments is to simulate the application of DRS to detect daily life scenarios of people working in the office or sleeping in bed.

In the first experimental scenario (the subject is a 25-year-old male with a height of 173 cm and a weight of 70 kg), the subject's arm is placed on a 3-D-printed holder with a hole designed for the radar to detect the wrist position, as shown in Fig. 9(a). In the second scenario, the same subject is lying on a bed where part of the mattress can be removed for the radar to detect the wrist position, as shown in Fig. 9(b). The distance from the radar antenna to the wrist position in these two experiments is set to be around 15 cm.

To verify the accuracy of the detected DWP, a piezoresistive pulse sensor, HK-2000 (Huake Information Technology, China) is placed on the same arm to detect the wrist pulse as a reference simultaneously. The output signal from the piezoresistive sensor and the I/Q signal from the DRS can be sampled synchronously by the AD7606 (Analog Device, MA, USA), which can be more accurate for comparison. Fig. 10 shows the subject's reference wrist pulse wave, the extracted DWP, and the correlated spectrum in different scenarios. Compared to the sitting position, the respiration and the corresponding harmonic components have less influence on the wrist pulse in the lying position, as shown in Fig. 10(b) and (d). From the frequency spectrum, it can be seen that the proposed respiratory and harmonics index K can work well for the respiratory signal. The green dashed lines are plotted to show the position of each systolic peak. The DWP waveform extracted by the ARHF method has a high agreement with

the waveform obtained by the reference contact sensor. It is worth noting that in both scenarios, the harmonic of the pulse beat can be found in the spectral domain, which shows the reliability of the designed DRS.

D. rPTT Measurement Analysis

The effectiveness of the proposed method in extracting accurate pulse waveforms was initially evaluated through a comparative analysis of rPTT measurements between the proposed and traditional filtering approaches. The rPTT is defined as the time duration from systolic peak to diastolic peak, as shown in Fig. 1. Fig. 11 shows the comparison of the extracted rPTT between the obtained DWP and the reference sensor. The evaluation metrics are selected as mean error (ME), root-mean-square error (RMSE), and normalized RMSE (NRMSE). The mathematical expressions for these metrics are

$$\begin{aligned} \text{ME} &= \frac{1}{n} \sum_{i=1}^n (X_i) = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i) \\ \text{RMSE} &= \sqrt{\frac{\sum_{i=1}^n (x_i - \hat{x}_i)^2}{n}} \\ \text{NRMSE} &= \frac{\text{RMSE}}{\bar{x}} = \frac{\sqrt{\frac{\sum_{i=1}^n (x_i - \hat{x}_i)^2}{n}}}{\bar{x}} \end{aligned} \quad (16)$$

where X is the difference between the estimated and the reference values. $\hat{x}_i$, x_i , and $\bar{x}$ are the estimated, the reference, and the average of reference rPTT values, respectively. The experimental results show that for participant 1# in Section C, the proposed method achieved ME, RMSE, and NRMSE values of 9.46 ms, 19.88 ms, and 0.34%, respectively, in the sitting position and 5.25 ms, 10.75 ms, and 0.19% in the lying position. In addition, to verify the reproducibility of the proposed method, the same experiments were performed on seven more subjects, with results summarized in Tables II and III. These results demonstrate that the proposed method consistently outperforms traditional approaches in rPTT measurement accuracy across different subjects and postures.

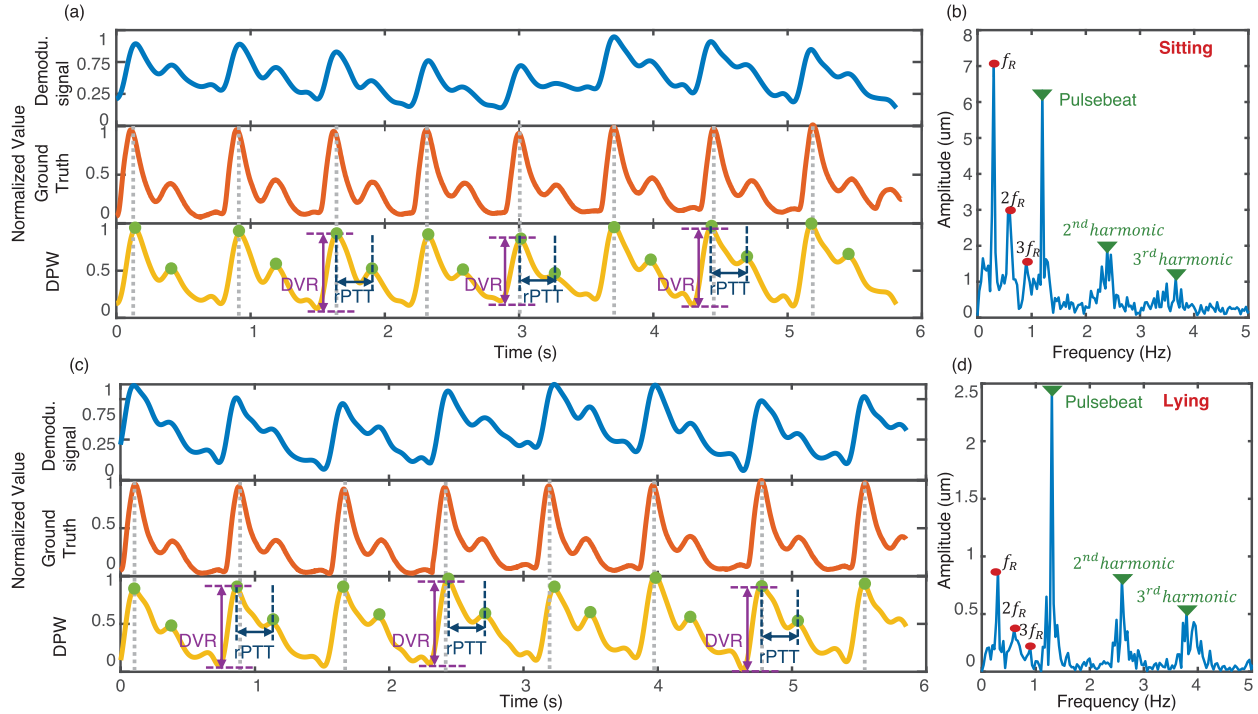


Fig. 10. Experimental results in two scenarios: the normalized synchronized recorded signal of the subject in (a) sitting and (c) lying position. From the top row to the bottom row, each row represents the demodulated signal (Demodu. signal), ground truth, and DWP, respectively. The frequency-domain representations of the demodulated signals from the radar sensor in setups (a) and (c) are shown in (b) and (d), respectively. In both cases, the value of K is 3. The gray dashed line represents the time stamp of each systolic peak in the ground-truth waveform.

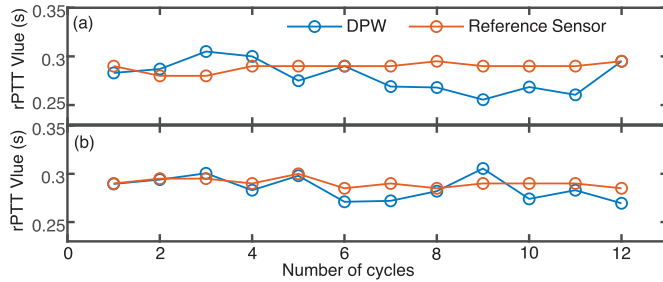


Fig. 11. rPTT comparison with the proposed method in different setups: subject in (a) sitting and (b) lying position.

TABLE I

PARTICIPANT INFORMATION INVOLVED IN THE EXPERIMENT

Subject	Gender	Health type	Age	Weight (kg)	Height (cm)
1#	Male	Healthy	25	70	173
2#	Male	Healthy	26	68	170
3#	Male	Healthy	32	75	180
4#	Female	Healthy	18	50	166
5#	Female	Healthy	24	52	168
6#	Female	Healthy	31	53	170
7#	Male	Hypertension	55	73	171
8#	Male	Hypertension	53	70	169

E. Contactless Blood Pressure Estimation

Once the DWP signal is detected, the contactless BP can be calculated by the proposed method. In the experiment, the participant was first asked to remain calm for 10 min to reduce emotional affect and other confounding factors. Then,

TABLE II

RPTT COMPARISON RESULT WHEN THE PARTICIPANTS ARE SITTING

Method	This Work			CEEMDAN			IIR with Comp.		
	ME (ms)	RMSE (ms)	NRMSE	ME (ms)	RMSE (ms)	NRMSE	ME (ms)	RMSE (ms)	NRMSE
Subject									
1#	9.46	19.88	0.34%	14.45	32.80	0.56%	11.00	21.25	0.36%
2#	-7.56	24.75	0.47%	-15.40	47.30	0.89%	-12.89	32.32	0.61%
3#	1.33	17.79	0.30%	14.20	34.55	0.58%	7.30	26.10	0.44%
4#	2.63	8.84	0.19%	17.10	30.15	0.61%	5.45	18.75	0.38%
5#	18.50	16.50	0.33%	19.25	33.75	0.68%	18.75	26.15	0.53%
6#	5.29	17.85	0.35%	11.00	36.75	0.73%	7.60	17.75	0.37%
7#	10.25	24.29	0.44%	-18.40	33.90	0.61%	12.25	26.15	0.47%
8#	13.92	24.24	0.41%	15.35	24.05	0.41%	15.05	24.50	0.41%

TABLE III

RPTT COMPARISON RESULT WHEN THE PARTICIPANTS ARE LYING

Method	This Work			CEEMDAN			IIR with Comp.		
	ME (ms)	RMSE (ms)	NRMSE	ME (ms)	RMSE (ms)	NRMSE	ME (ms)	RMSE (ms)	NRMSE
Subject									
1#	5.25	10.75	0.19%	13.65	39.05	0.68%	6.88	19.65	0.34%
2#	8.17	15.71	0.32%	-11.76	37.55	0.77%	-10.04	26.80	0.54%
3#	6.17	17.49	0.30%	15.25	28.35	0.48%	6.40	24.25	0.41%
4#	3.70	19.72	0.40%	17.30	25.75	0.54%	4.46	18.55	0.39%
5#	8.08	11.86	0.24%	13.50	24.65	0.50%	13.30	21.85	0.44%
6#	4.21	8.64	0.18%	7.64	27.30	0.56%	9.70	23.95	0.48%
7#	14.79	22.60	0.38%	-17.60	25.10	0.43%	11.63	24.65	0.42%
8#	11.30	21.90	0.37%	16.60	31.05	0.53%	16.15	25.80	0.44%

the customized DRS collected the DWP for 30 s, accordingly, a medical digital sphygmomanometer, HEM-2010 (OMRON, Kyoto, Japan), was placed on the same arm to collect the blood pressure. To verify the stability of the proposed method and to reduce potential interferences, eight subjects participated in this experiment and the data were collected simultaneously for three consecutive days. Fig. 10 illustrates the characteristic

TABLE IV
COMPARISON BETWEEN TWO METHODS AND THE RESULT OF ALL SUBJECTS WITH SITTING POSTURE

Subject	DBP		SBP	
	ME $\pm$ SD (mmHg)		ME $\pm$ SD (mmHg)	
	Proposed method	Method by [22]	Proposed method	Method by [22]
1#	2.18 $\pm$ 0.45	2.21 $\pm$ 2.77	3.23 $\pm$ 2.37	4.87 $\pm$ 3.99
2#	-0.64 $\pm$ 0.88	-1.13 $\pm$ 4.41	8.38 $\pm$ 8.19	11.54 $\pm$ 15.14
3#	2.06 $\pm$ 2.37	4.24 $\pm$ 6.80	-2.56 $\pm$ 6.17	4.44 $\pm$ 4.84
4#	-1.32 $\pm$ 1.78	2.31 $\pm$ 6.42	3.38 $\pm$ 9.54	-3.52 $\pm$ 15.67
5#	1.83 $\pm$ 4.93	5.72 $\pm$ 6.82	4.31 $\pm$ 8.54	3.76 $\pm$ 12.46
6#	0.96 $\pm$ 4.02	-1.95 $\pm$ 5.24	2.50 $\pm$ 8.46	4.89 $\pm$ 13.74
7#	1.96 $\pm$ 5.31	-2.63 $\pm$ 6.88	-2.53 $\pm$ 8.89	-6.67 $\pm$ 21.23
8#	1.08 $\pm$ 3.13	1.37 $\pm$ 5.52	4.14 $\pm$ 12.58	7.91 $\pm$ 18.38
Total	1.01 $\pm$ 2.84	1.27 $\pm$ 6.33	2.60 $\pm$ 7.89	3.40 $\pm$ 10.31

TABLE V
COMPARISON BETWEEN TWO METHODS AND THE RESULT OF ALL SUBJECTS WITH LYING POSTURE

Subject	DBP		SBP	
	ME $\pm$ SD (mmHg)		ME $\pm$ SD (mmHg)	
	Proposed method	Method by [22]	Proposed method	Method by [22]
1#	2.60 $\pm$ 5.37	2.88 $\pm$ 9.75	8.68 $\pm$ 6.64	12.35 $\pm$ 6.50
2#	-1.54 $\pm$ 3.45	4.21 $\pm$ 5.74	8.92 $\pm$ 7.54	10.91 $\pm$ 8.14
3#	2.93 $\pm$ 1.94	1.06 $\pm$ 6.09	5.86 $\pm$ 8.05	6.57 $\pm$ 7.79
4#	1.79 $\pm$ 2.79	1.98 $\pm$ 3.42	-3.80 $\pm$ 15.83	-4.90 $\pm$ 18.40
5#	2.86 $\pm$ 5.54	2.52 $\pm$ 7.24	2.69 $\pm$ 9.86	3.17 $\pm$ 15.81
6#	-2.06 $\pm$ 3.39	-2.26 $\pm$ 9.27	4.59 $\pm$ 9.67	3.42 $\pm$ 10.90
7#	3.83 $\pm$ 3.96	4.26 $\pm$ 4.84	7.71 $\pm$ 20.04	6.87 $\pm$ 25.12
8#	1.83 $\pm$ 6.39	2.92 $\pm$ 8.62	5.01 $\pm$ 13.81	6.55 $\pm$ 18.31
Total	1.53 $\pm$ 4.68	2.20 $\pm$ 6.83	4.96 $\pm$ 10.15	5.62 $\pm$ 12.31

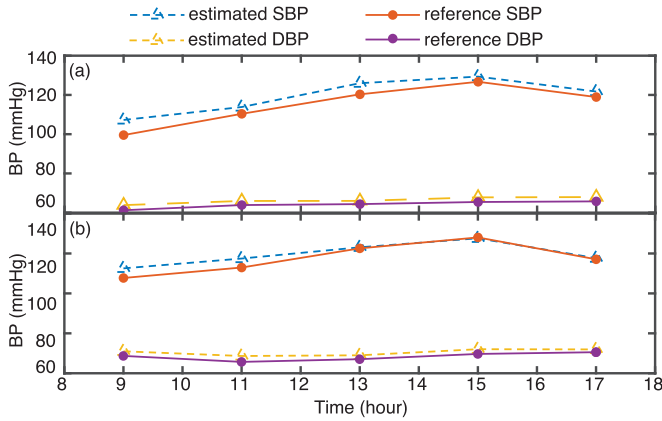


Fig. 12. Estimated and the reference BP values of subject 1 in sitting posture over two days. (a) Day 2. (b) Day 3.

points of the DWP signal, which were automatically identified using MATLAB, and the subsequent calculations of DVR and rPTT derived from these points. For clarity, only a few cycles of DVR and rPTT extraction are shown, in practice, this process is applied to all cycles. The data from the first day were used for calibration. It is worth noting that each calibration coefficient is different.

Fig. 12 shows the subject 1#’s estimated BP and the reference BP values in two days, demonstrating the overall performance of the proposed method. To further validate the proposed method, this study compared with recent works that did not utilize DVR for BP prediction [22]. The comparison results are included in Tables IV and V. The evaluation metrics of the proposed method are ME [the definition can be found in (16)] and standard deviation (SD), where the expression is

given as

$$\begin{aligned}
 SD &= \sqrt{\frac{\sum_{n=1}^i (X_i - \mu)^2}{n}} \\
 &= \sqrt{\frac{\sum_{n=1}^i ((x_i - \hat{x}_i) - \mu)^2}{n}}
 \end{aligned} \quad (17)$$

where X_i is the difference between the estimated and the reference values. $\hat{x}_i$ and x_i are the estimated and the reference BP values, respectively. μ is the mean of X . n denotes the number of samples. The calculation results are shown in Tables IV and V. In the sitting posture, the total average ME and SD for the proposed method are 1.01 ± 2.84 and 2.60 ± 7.89 mmHg for DBP and SBP, respectively. The total average ME and SD of DBP and SBP in lying posture are 1.53 ± 4.68 and 4.96 ± 10.15 mmHg, respectively. When compared with the method in [22], the proposed method shows improvements in both accuracy and consistency, with differences of 0.26 ± 3.49 mmHg for DBP and 0.80 ± 2.42 mmHg for SBP in the sitting posture.

To further assess the reliability of the proposed method, the Bland–Altman plots of SBP and DBP’s are applied to the first subject, which can be found in Fig. 13. All the estimated SBP and DBP values are within $ME \pm 1.96 SD$, which is the 95% confidence interval. These metrics show that the proposed BP estimation method can be reliable for the contactless BP field.

The proposed method demonstrates better stability across different subjects and postures, while the method in [22] shows larger fluctuations in some cases (e.g., subject 7# in SBP estimation). It is worth noting that individual differences exist among subjects, and the relationship between hypertension status and measurement variability appears to be

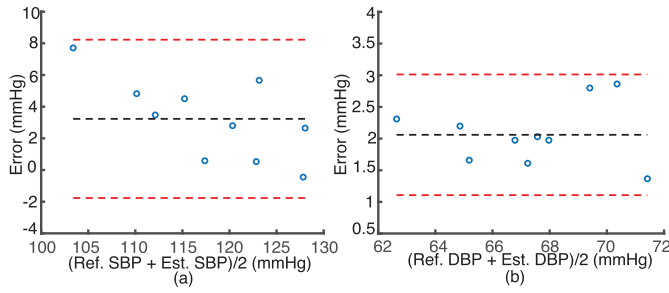


Fig. 13. Bland–Altman plots of (a) SBP value and (b) DBP value of the first subject. The dark dashed line represents the mean error and the red dashed line represents the mean $\pm 1.96SD$.

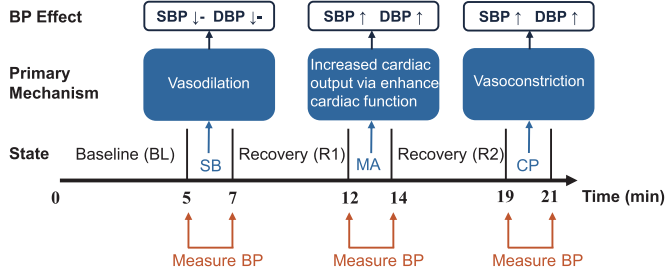


Fig. 14. Characteristic points of the wrist pulse wave.

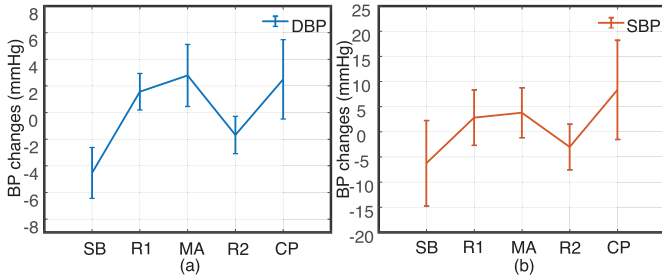


Fig. 15. Total average result of BP comparison in five motions. (a) DBP difference variations in five motions. (b) SBP difference variations in five motions. The value of each point represents ME $\pm$ SD.

subject-specific. For instance, subjects 7# and 8#, while classified as hypertensive, do not consistently show higher variability or significantly different patterns compared to other subjects. Overall, DBP estimation is more accurate than SBP estimation in both postures. This reflects the challenges in SBP estimation and highlights an area for future research focus.

F. Specific BP Fluctuation Range Estimation

To test the robustness of the proposed method for continuous BP measurement in various states, experiments were conducted healthy subjects to induce fluctuations in BP, including scenarios involving slow breathing (SB) [37], mental arithmetic (MA) [38], cold pressor (CP) [39], and recovery periods (R1 and R2) [40]. Fig. 14 records the detailed time arrangement. BP measurements were taken at the end of each rest stage and movement, and volunteers immediately acquired 30 s of radar DWP. The training data for each volunteer will consist of the baseline BP values and related DWP parameters. Fig. 15 displays the average DBP and SBP estimation results of all volunteers under different exercise states, with the center

TABLE VI
COMPARISON WITH OTHER SENSORS

Sensor type	Contact or not	Measurement type	Sensitive to environment	Privacy issue
Sphygmomanometer	Contact	Discontinuous	Low	No
Cameras	Contactless	Continuous	High	Yes
PPG sensors	Contact	Continuous	Medium	No
Piezoelectric sensors	Contact	Continuous	Medium	No
Radar	Mostly contactless	Continuous	Low	No

representing the ME and the vertical bars representing the SD, respectively.

Fig. 15 indicates that the predicted values of DBP and SBP are negative during the SB period, suggesting that they surpass the readings from the reference medical digital sphygmomanometer. This phenomenon may be attributed to the relatively high readings during the BL, leading to an overestimation of the overall predicted values by the model. Nonetheless, the proposed method exhibits commendable predictive accuracy when specific activities induce BP elevation, such as during MA and CP, in comparison to the SB period. The highest inaccuracies in DBP and SBP both occur in the CP periods, resulting in errors of 2.50 ± 2.98 and 8.34 ± 9.86 mmHg, respectively. It is worth noting that the SD predicted by DBP is smaller than that of SBP, indicating a necessity for further research to minimize SD error in predicting SBP.

G. Summary

This section includes both DRS and practical experiments. Initially, the experiment validates the radar's high linearity and sensitivity, with a minimum detection accuracy of $2 \mu\text{m}$. Subsequently, the proposed ARHF method accurately extracts the DWP from respiration and its harmonic components, achieving minimal rPTT inaccuracy. Although the intensity of respiratory components during sitting is higher than that during lying, the combination of the proposed ARHF method and high-gain DRS can effectively capture wrist motion. With this method, DWP signals can be effectively extracted from wrist movements containing respiration and its harmonics. It should be noted that the differences in DWP signals extracted from distinct subjects have led to different measurement comparisons of rPTT values.

The proposed ARHF method provides a fundamental framework for contactless BP estimation. Subsequently, an experiment on stationary subject BP estimation was carried out. The experiment result demonstrates that the proposed BP estimation method improves the accuracy of BP estimation compared to the conventional rPTT method proposed by [22]. Furthermore, based on the Bland–Altman plots, the contactless method for estimating BP demonstrates reliability with subjects without using a cuff or contact probes.

To validate the efficacy and robustness of this method in varied BP ranges, an additional specific BP fluctuation range estimation experiment was conducted, with healthy volunteers participating in the exercise that induced fluctuations in BP readings. The experimental results validate the robustness of the proposed method and indicate that the proposed method

TABLE VII
COMPARISON BETWEEN OTHER RADAR-BASED METHODS. S: SITTING POSTURE. L: LYING POSTURE

Work	Detection site	Contact or not	Working mode and frequency	Antenna gain	Filtering method	Respiratory and harmonics suppression	Accuracy
[8]	Wrist	Contact	CW 5.7 GHz	4.7 dBi	IIR	No	(ME $\pm$ SD) mmHg SBP: -3.79 $\pm$ 6.32 DBP: -3.4 $\pm$ 6.23
[9]	Wrist and chest	Contactless	CW 2.4 GHz	14 dBi	Empirical mode decomposition	Yes	(MAE $\pm$ SD) mmHg SBP: 9.51 $\pm$ 12.13 DBP: 6.69 $\pm$ 8.12
[10]	Wrist and chest	Contactless	CW 5.8 & 120 GHz	6.2 dBi (120 GHz) 8 dBi	FIR	No	Mean relative error <3%
[22]	Wrist	Contactless	CW 120 GHz	8 dBi	FIR	No	mmHg SBP < 2.8 DBP < 2.2
[42]	Chest	Contactless	CW 300 GHz	15 dBi	Discrete wavelet transform	Yes	SBP: 8.04% DBP: 8.3%
This work	Wrist	Contactless	CW 24 GHz	17.3 dBi	ARHF	Yes	(ME $\pm$ SD) mmHg S: SBP: 2.60 $\pm$ 7.89 DBP: 1.01 $\pm$ 2.84 L: SBP: 4.96 $\pm$ 10.15 DBP: 1.53 $\pm$ 4.68

can realize continuous BP measurement. This indicates that our work is not limited to technical validation but also focuses on the robustness of practical applications and methods, providing useful references and directions for future in-depth research.

IV. DISCUSSION

A. Comparison With Other Noninvasive Sensors

Table VI compares different sensors utilized for BP estimation. The sphygmomanometer is commonly regarded as the gold standard for noninvasive BP measurement. However, it can be uncomfortable for the user due to the pressure applied to the arm during measurement. PPG and piezoelectric sensors estimate BP by capturing the pulse waveform without requiring substantial pressure on the skin. Nonetheless, both sensors necessitate close contact with the skin, which may still cause discomfort and have limited battery life, which prevents prolonged monitoring. Additionally, PPG is highly susceptible to environmental interference such as sweat [41], which affects measurement accuracy. Contactless sensors like cameras face issues of light sensitivity and privacy concerns, making them less suitable for BP measurements. The radar system utilized in this study effectively addresses these challenges, enabling contactless measurements and accurate extraction of DWP for BP estimation when combined with the ARHF algorithm. Nonetheless, further detailed experiments are needed to confirm the broad applicability of the proposed system.

B. Comparison With Other Radar-Based Works

Table VII lists the parameters involved in recent radar-based studies for BP estimation. These parameters include the detection site, sensing method, working mode and frequency, antenna gain, filtering technique, and the results. The studies have mainly focused on the wrist or the chest. In this work, choosing the wrist as the detection location is important because the signals from breathing and its harmonics are weaker than those at the upper part of the body, making the wrist pulse signal less susceptible to interference.

However, other detecting wrist pulses studies, such as [8], [10], [22], have not thoroughly explored the effect of respiratory signals at the wrist. This lack of exploration leads to interference in extracting DWP, resulting in degraded system performance. Furthermore, studies employing alternative approaches [9], [42] may introduce nonlinearities through filtering methods [43], potentially causing distortions in weak signals. This distortion could affect the accurate detection of DWP, leading to an overall degradation in performance. Additionally, compared to [8], the proposed method has a smaller SD, implying more stable predicted fluctuations.

C. Comparison With Different Postures

Tables IV and V display the BP measurements of the subjects in sitting and lying positions, respectively. The experiments reveal that the overall errors of DBP and SBP are lower when the experimenter is sitting. The estimation error can be attributed to two factors: First, the prediction model itself may have a certain error. Second, experimental limitations exist, particularly regarding the sphygmomanometer's requirement for subjects to maintain a sitting position for accurate measurements [44]. For measurements in the lying position, rapid postural transitions are required because DWP needs to be obtained in a short time. Due to individual physiological variations among subjects, these postural transitions can induce additional BP fluctuations. Furthermore, the radar system predicts the BP value of the user in the lying position, not the BP value in the sitting position. Consequently, these factors may lead to increased BP estimation errors in the lying position.

D. Limitations of This Work

The work presented in this article still has some limitations. Although radar-based BP monitoring shows promise, it struggles to effectively deal with motion artifacts. To minimize arm movement, we designed a 3-D-printed box to support the arm. However, there are limitations to its practical application. Additionally, the detection distance can impact the results, as DWP waveforms beyond a certain range are not well

detected, affecting their generalizability for practical applications. These issues will be the focus of future research.

V. CONCLUSION

This study introduces a continuous contactless BP estimation technique based on pulse intensity ratio, which relies on the DWP signal detected by customized high-gain DRS. Initially, the experiment validates the radar's high linearity and sensitivity. The proposed ARHF method accurately extracts DWP with minimal rPTT inaccuracy. In comparison to existing methods, the proposed continuous contactless BP estimation model improves prediction accuracy. Specific BP fluctuation range changes experiment and predictions validate the reliability and robustness of the proposed method. Overall, the methodologies proposed in this work enhance the utility of DRS for a wide array of healthcare applications, highlighting the important contribution to improving BP monitoring and prediction. Future work could explore higher working frequency, with more precise signals, to improve BP estimation accuracy and enhance robustness across different age groups.

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